

Data Warehousing (DW)

Lecture-1: Introductory lecture

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Course Objectives

- What is Data Warehousing (DW), its main characteristics.
- Show how data is integrated from multiple data sources into a single data repository (DW).
- DW Design Approaches and Choices.
- Three Tiers of DW Lifecycle.
- The DW pipeline.
- Multidimensional Data Visualization.
- DW Indexing

What is a Data Warehouse

- **A DW is a:**

- Subject Oriented,
- Integrated,
- Time Variant
- Nonvolatile

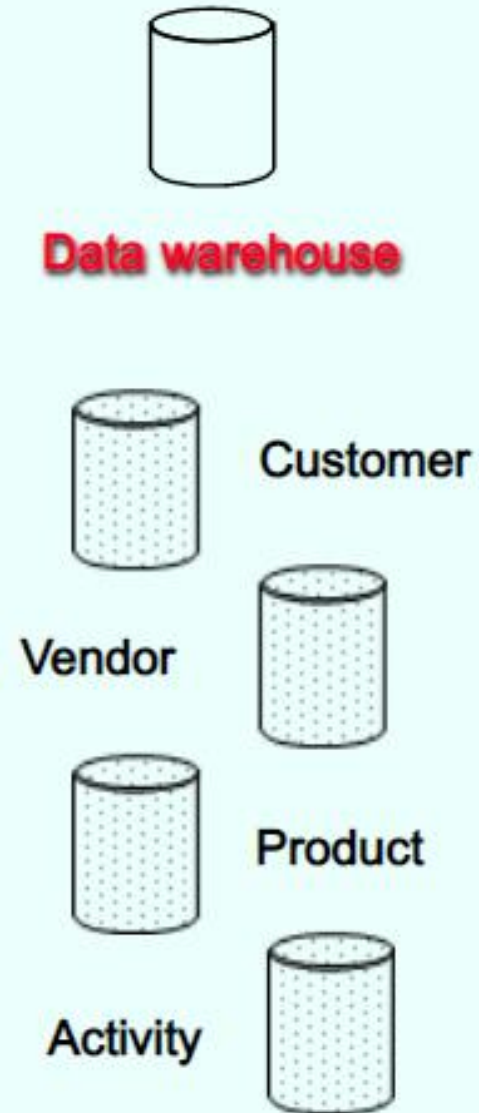
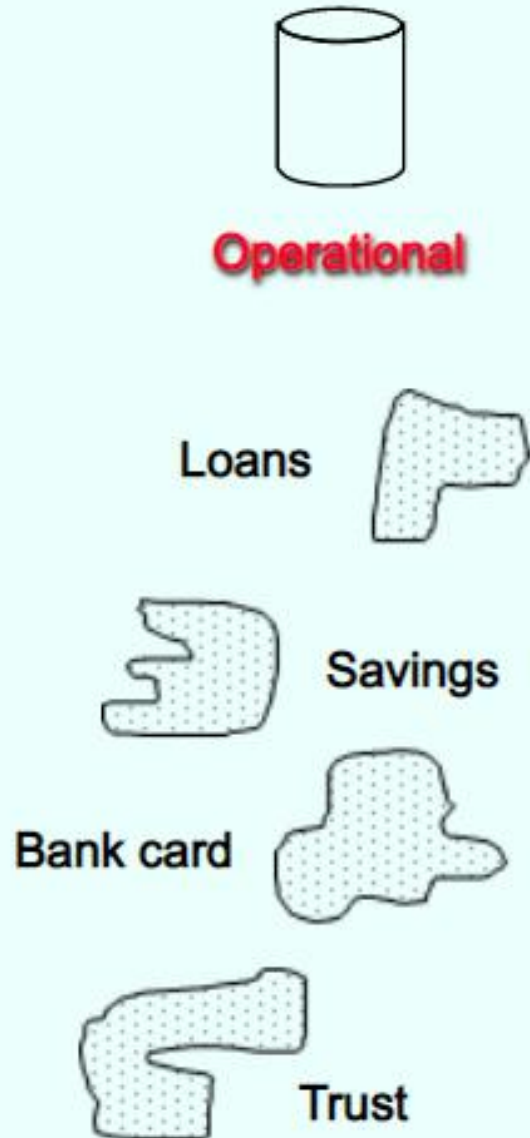
Collection of data in support of management's decision-making process.

- **The data sources:** Operational Environment.

What is a Data Warehousing

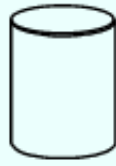
- **DW is oriented around the major subjects of the enterprise.**
- **DW excludes data that will not be used for DSS.**
- **DW spans a spectrum of time.**
- **It is technique that gathers or collects data from different sources into central repository.**
- **It is a powerful database model that enhances user ability to analyze huge multidimensional data sets to allow user to make a business decision based on facts and for tracking quick and effective decisions or providing necessary information.**

DW has a Strong Subject Orientation



Integration

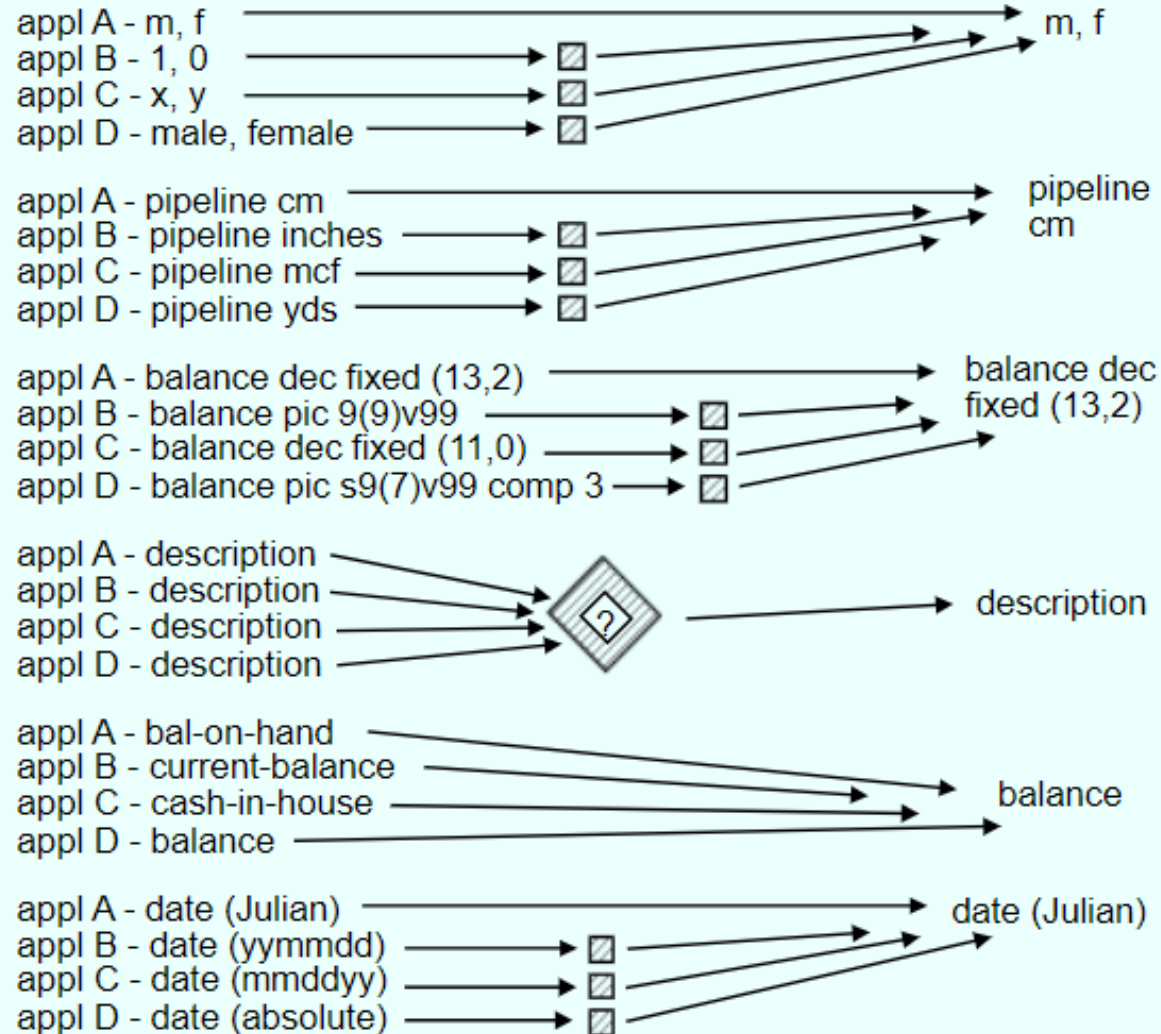
- Data found within the DW is integrated
 - ✓ Consistent naming conventions
 - ✓ Consistent measurement of variables
 - ✓ Consistent encoding structures.



Operational



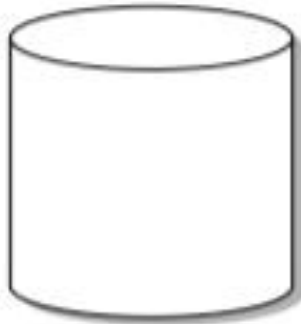
Data warehouse



Time Variance

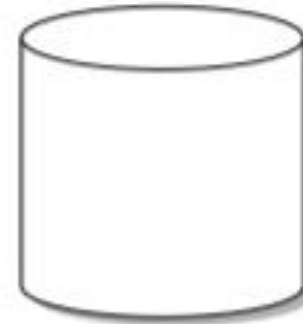
- All data in the data warehouse is accurate as of some moment in time (i.e., not "right now").
- In the operational environment data is accurate of access.
- Data found in the warehouse is said to be "time variant".

Operational



- Time horizon – current to 60–90 days
- Update of records
- Key structure may or may not contain an element of time

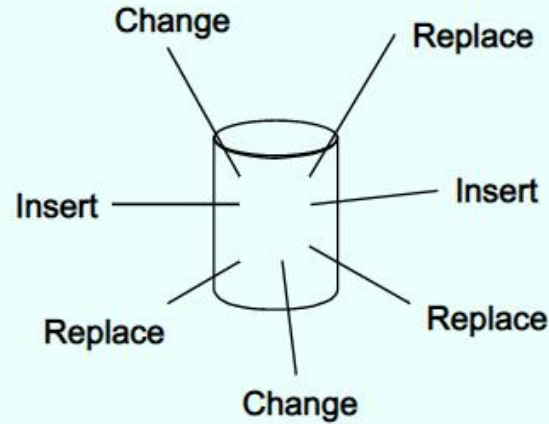
Data warehouse



- Time horizon – 5–10 years
- Sophisticated snapshots of data
- Key structure contains an element of time

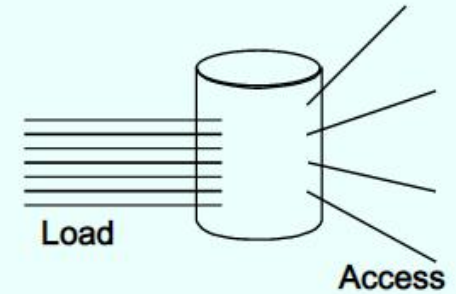
Non-Volatile

- The basic manipulation of data that occurs in the data warehouse is simple.
- There are only two kinds of operations
 - ✓ Loading:
 - ✓ The initial loading of data
 - ✓ Refreshment process
 - ✓ The access of data
- There is no update of data
- The need to be cautious of the update anomaly is no factor



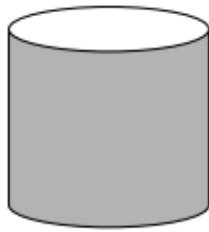
Operational

Data is updated on a record-by-record basis regularly



Data warehouse

Data is loaded into the warehouse and is accessed there, but once the snapshot of data is made, the data in the warehouse does not change



GRANULARITY– THE LEVEL OF DETAIL

High level of detail–
low level of granularity

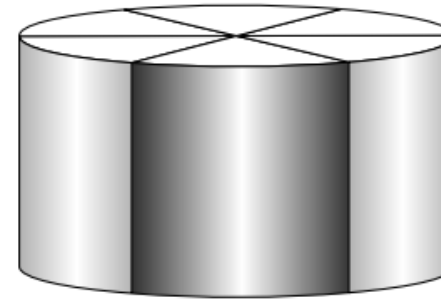
EXAMPLE:
The details of every
phone call made by a
customer for a month



Low level of detail–
high level of granularity

EXAMPLE:
The summary of phone
calls made by a
customer for a month

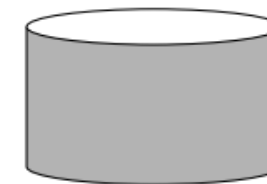
Granularity
Level



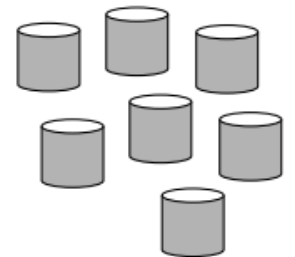
PARTITIONING OF DATA

- The splitting of data into small units
- Done at the application level or the DBMS level

(b)



Difficult to manage



Easy to manage

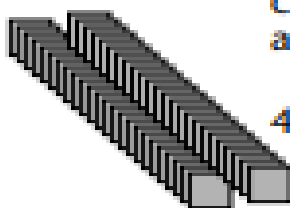
Example of granularities

GRANULARITY

High level of detail



EXAMPLE:
The details of every phone
call made by a customer for
a month



40,000 bytes per month
200 records per month

```
01 activityrec.  
  02 date  
  02 time  
  02 to whom  
  02 op assisted  
  02 call completed  
  02 time completed  
  02 long distance  
  02 cellular  
  02 special rate  
  .....  
  .....
```

Low level of detail



EXAMPLE:
The summary of phone
calls made by a customer
for a month

 200 bytes
1 record per month

```
01 activityrec.  
  02 month  
  02 cumcalls  
  02 avglength  
  02 cumlongdistance  
  02 cuminterrupted  
  .....
```

Granularity level controls type of queries to be answered

GRANULARITY

High level of detail



Example:
The details of every phone call made by a customer for a month

Low level of detail



Example:
The summary of phone calls made by a customer for a month

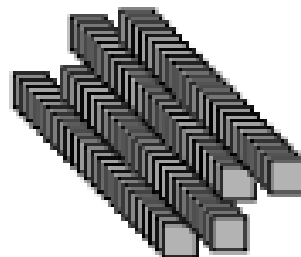
"Did Cass Squire call his girlfriend in Boston last week?"

- Can be answered, even though some amount of digging is required.

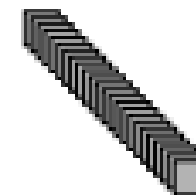
- Cannot be answered in any case. The detail has gone.

But looking for a single record is a very uncommon event.

"On the average, how many long-distance phone calls did people from Washington make last month?"



Search through 175,000,000 records, doing 45,000,000 I/Os



Search through 1,750,000 records, doing 450,000 I/Os

Data Warehousing Modeling

Warehouse Models & Operators

- **Data Models**
 - Relations
 - Stars (Star Schema) & Snowflakes
 - Cubes
- **Operators**
 - Slice & Dice
 - Roll-up & Drill Down
 - Pivoting
 - Other Operators.

Multi-Dimensional Data

- **Measures: Raw Facts & Derived Facts (Measures)**

- Numerical
- Additive

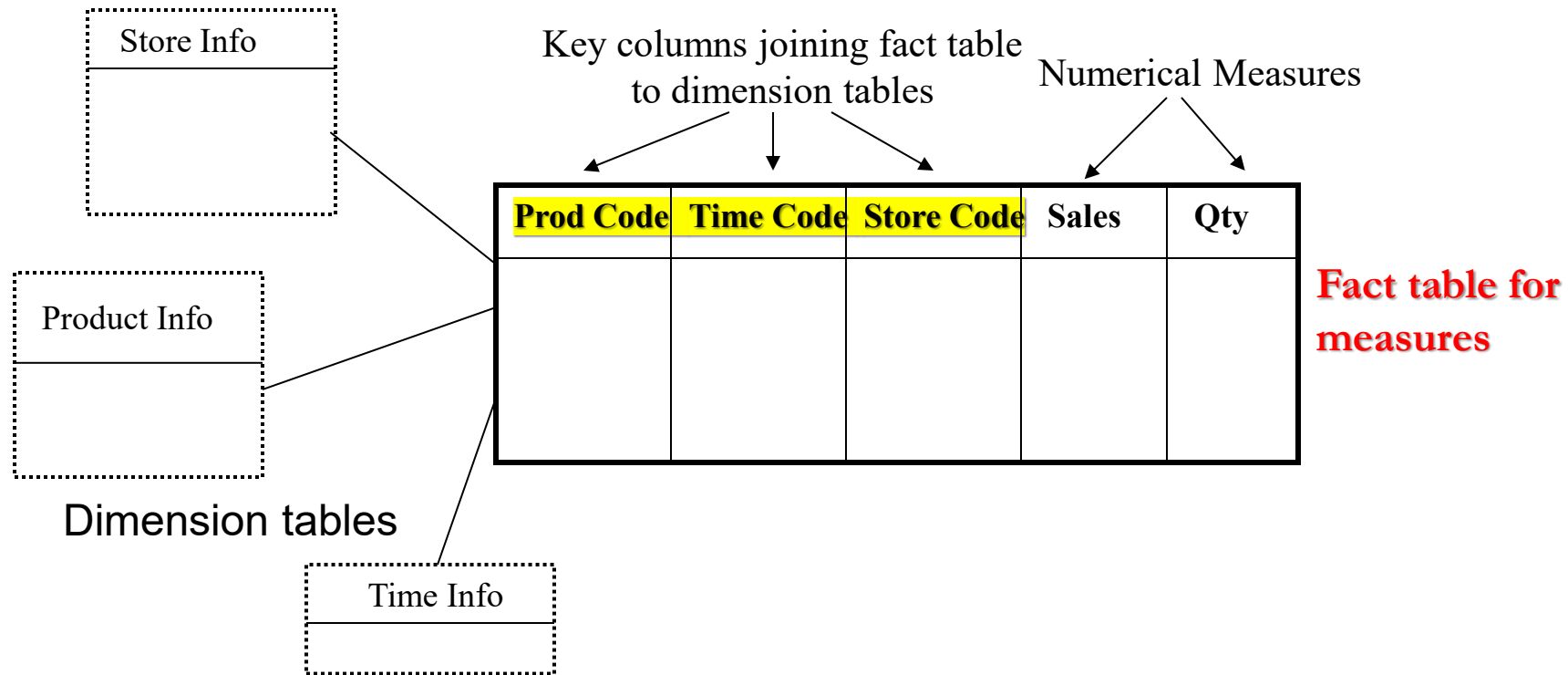
Data being tracked in business, can be analyzed & examined

- **Dimensions** - business parameters that define a transaction, relatively static data such as lookup or reference tables
- **Example:** Analyst may want to view sales data, by geography, by time, and by product

The Multi-Dimensional Model

“Sales by product line over the past six months”

“Sales by store between 1990 and 1995”



Multidimensional Modeling

- **Multi-dimensional Modeling** is a technique for structuring data around the business concepts
- ER models describe “**ENTITIES**” and “**RELATIONSHIPS**”
- **Multi-dimensional Models** describe “measures” and “dimensions”

Dimensional Modeling

- **Dimensions are organized into hierarchies**
 - E.g., Time dimension: days → weeks → quarters
 - E.g., Product dimension: product → product line → brand
- **Dimensions have attributes**

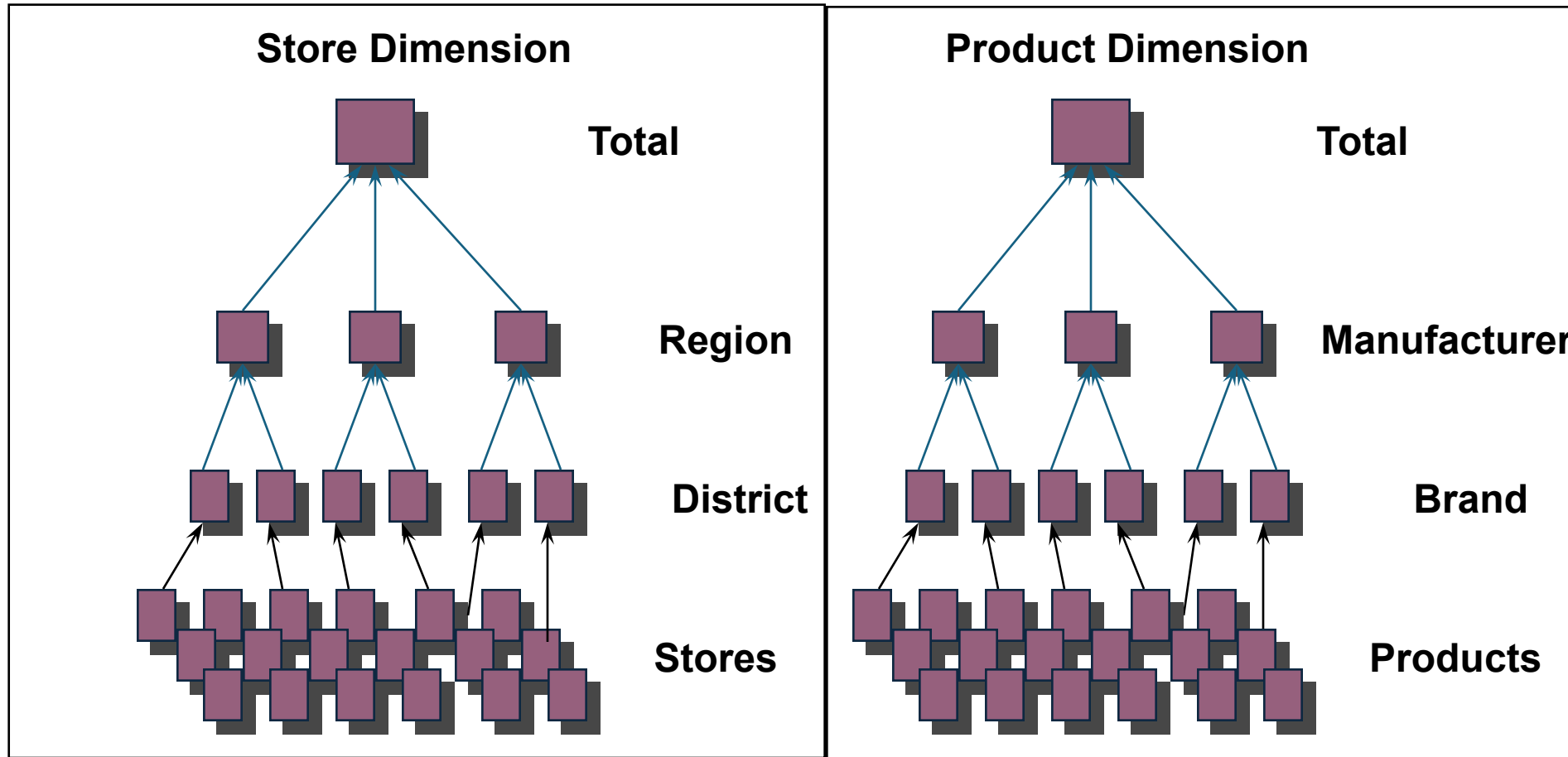
Time

Store

StoreID
City
State
Country
Region

Date
Month
Year

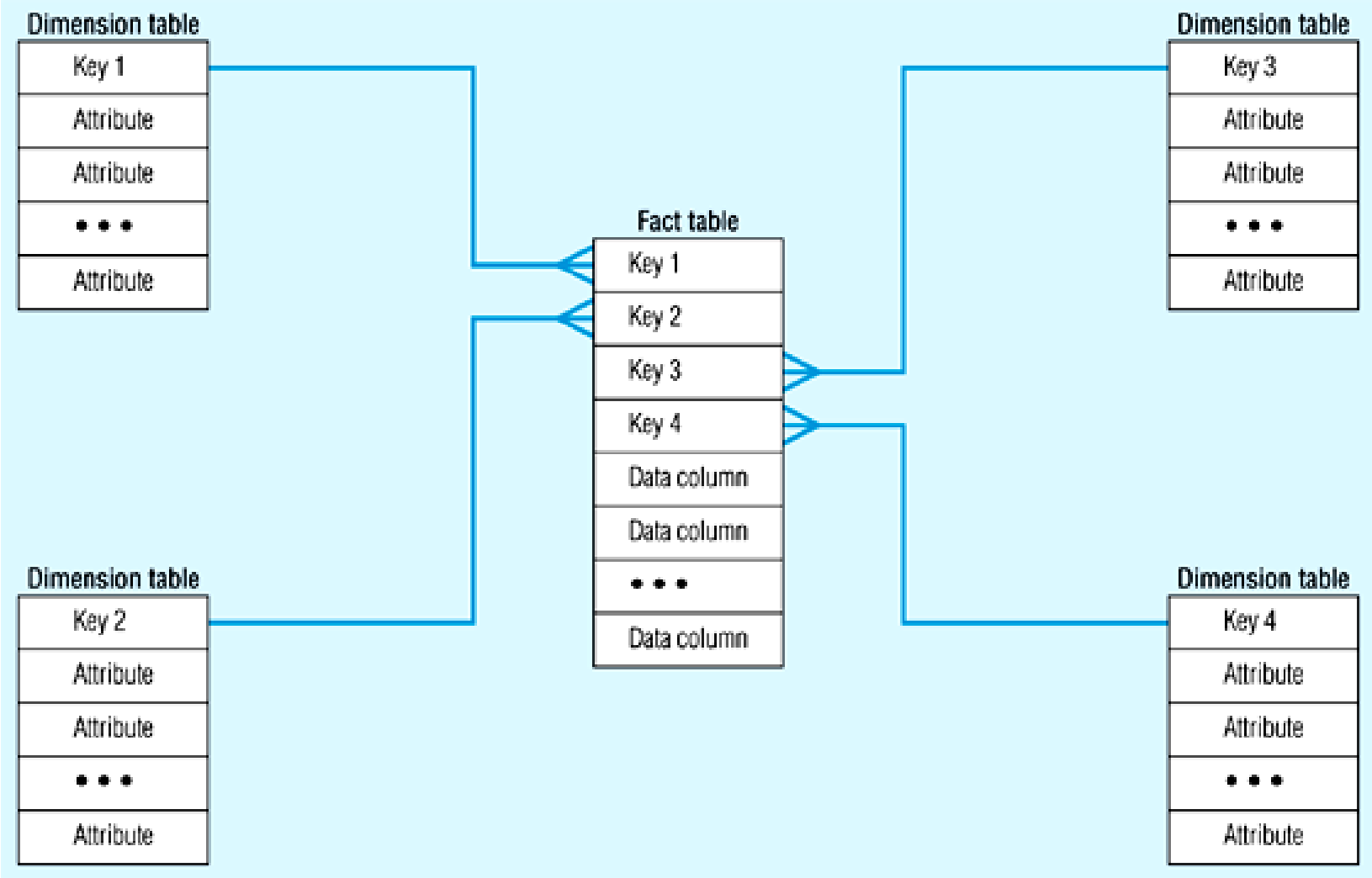
Dimension Hierarchies



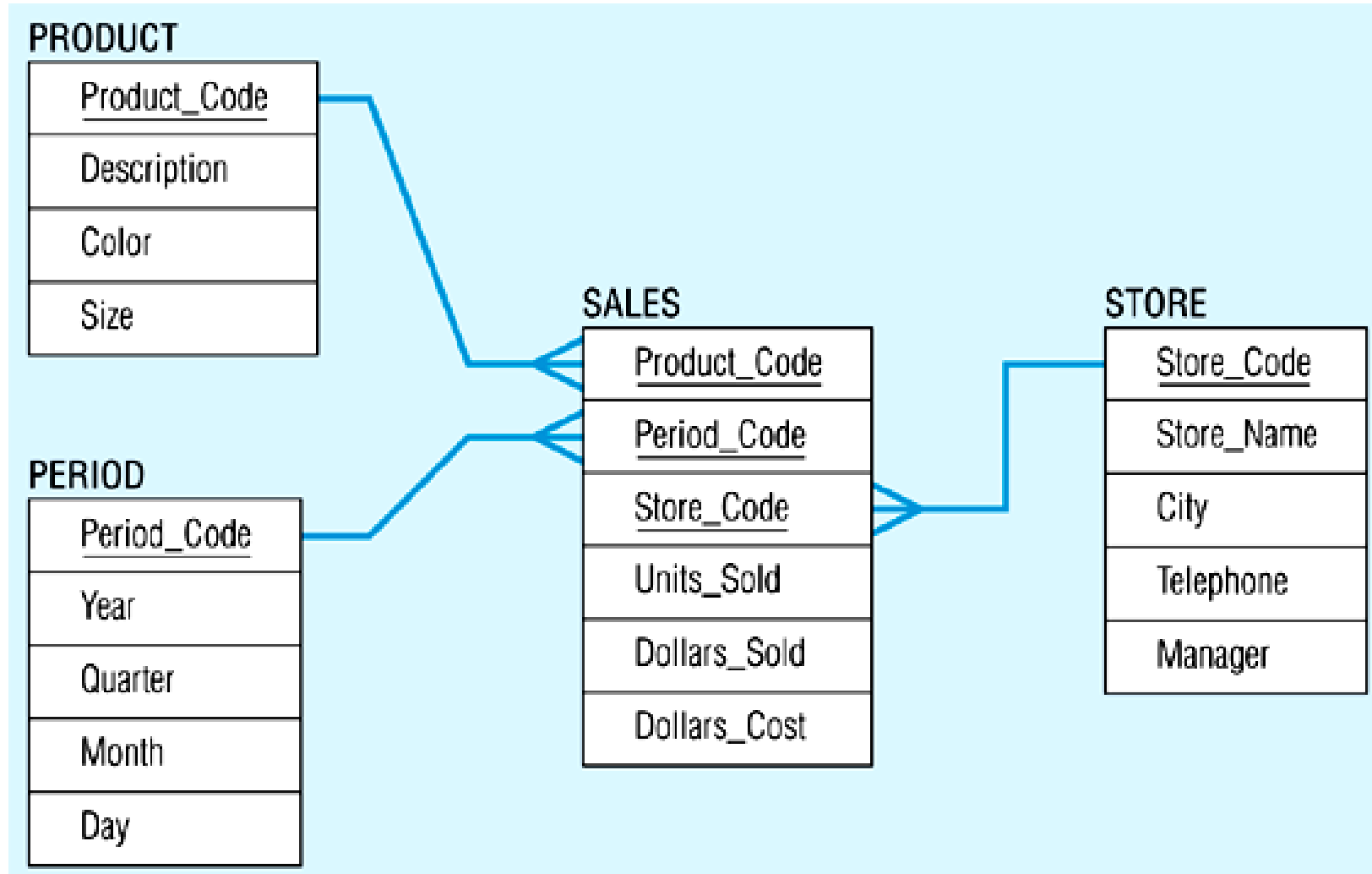
Schema Design

- Most data warehouses use a star schema to represent multi-dimensional model.
- Each dimension is represented by a **dimension table** that describes it.
- A **FACT TABLE** connects to all dimension tables with a multiple join.
 - Each tuple in the fact table consists of a pointer to each of the dimension tables that provide its multi-dimensional coordinates and stores measures for those coordinates.
 - The links between the fact table in the center and the dimension tables in the extremities form a shape like a **STAR**.

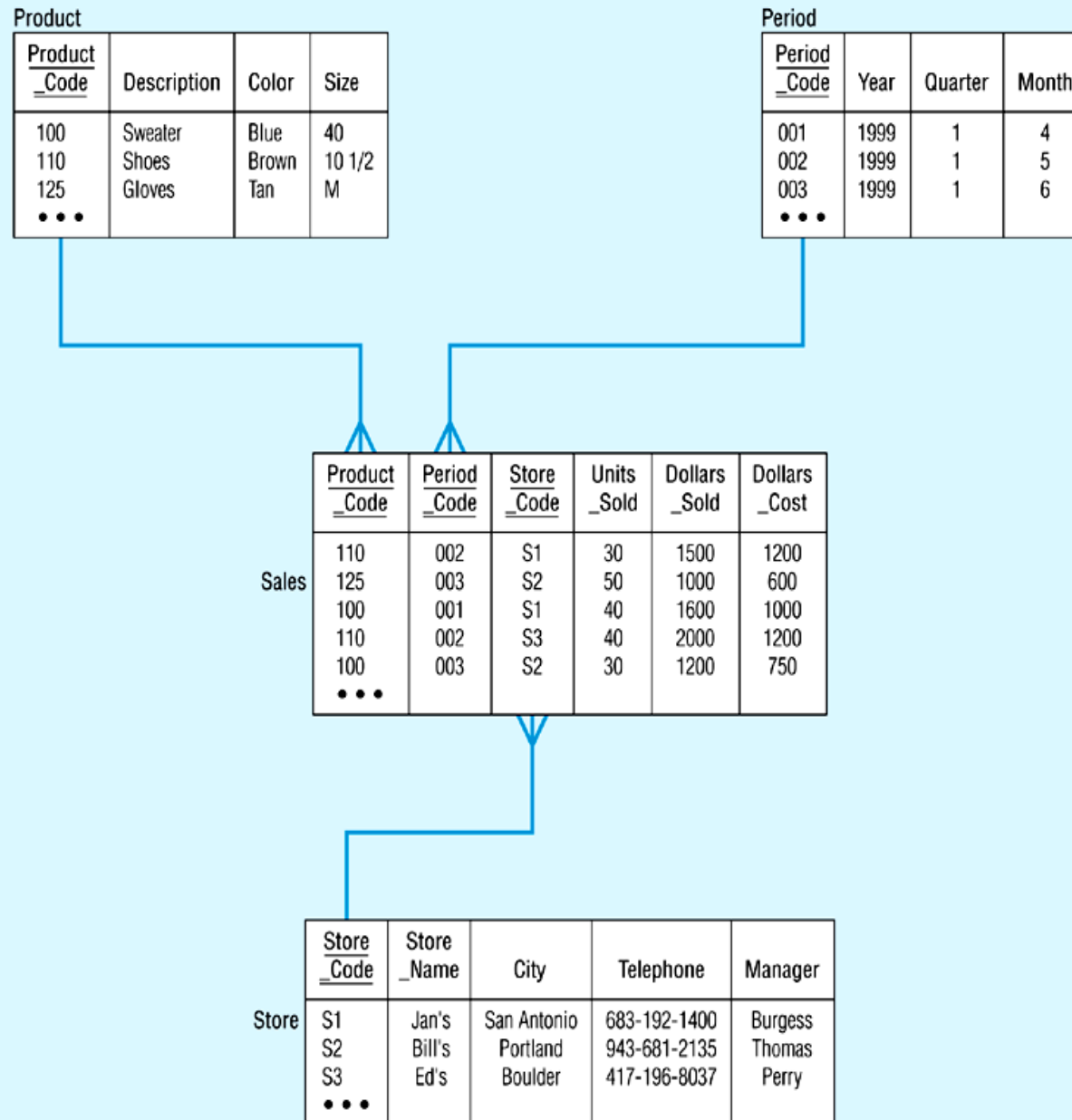
Star Schema (in RDBMS)



Star Schema Example



Star Schema with Sample Data



The “Classic” Star Schema

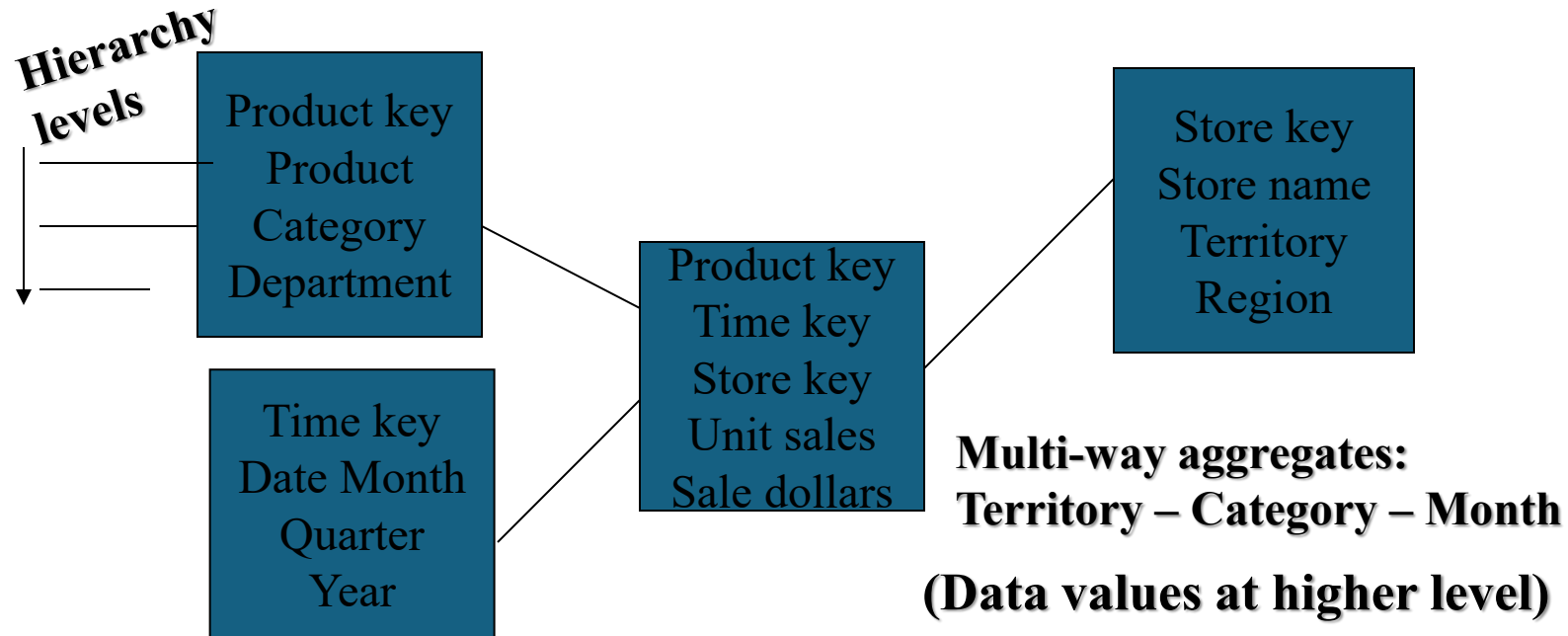
- ◆ A relational model with a one-to-many relationship between dimension table and fact table.
- ◆ A single fact table, with detail and summary data
- ◆ Fact table primary key has only one key column per dimension
- ◆ Each dimension is a single table, highly denormalized
- **Benefits:** Easy to understand, intuitive mapping between the business entities, easy to define hierarchies, reduces # of physical joins, low maintenance, very simple metadata
- **Drawbacks:** Summary data in the fact table yields poorer performance for summary levels, huge dimension tables a problem

Need for Aggregates

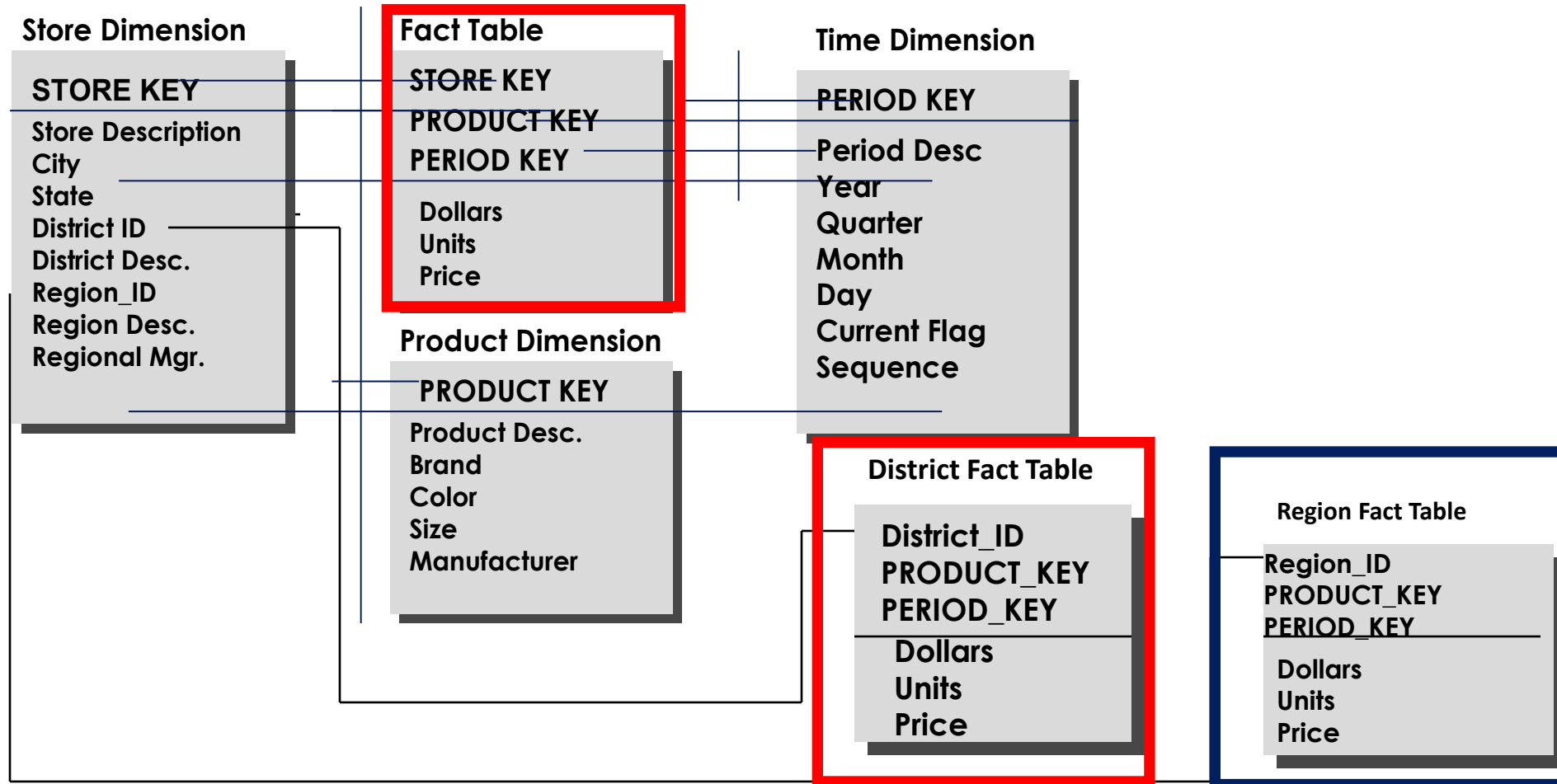
- Sizes of typical tables:
 - Time dimension: 5 years x 365 days = 1825
 - Store dimension: 300 stores reporting daily sales
 - Production dimension: 40,000 products in each store (about 4000 sell in each store daily)
 - Maximum number of base fact table records: 2 billion (lowest level of detail)
- A query involving 1 brand, all store, 1 year: retrieve/summarize over 7 million fact table rows.

Aggregating Fact Tables

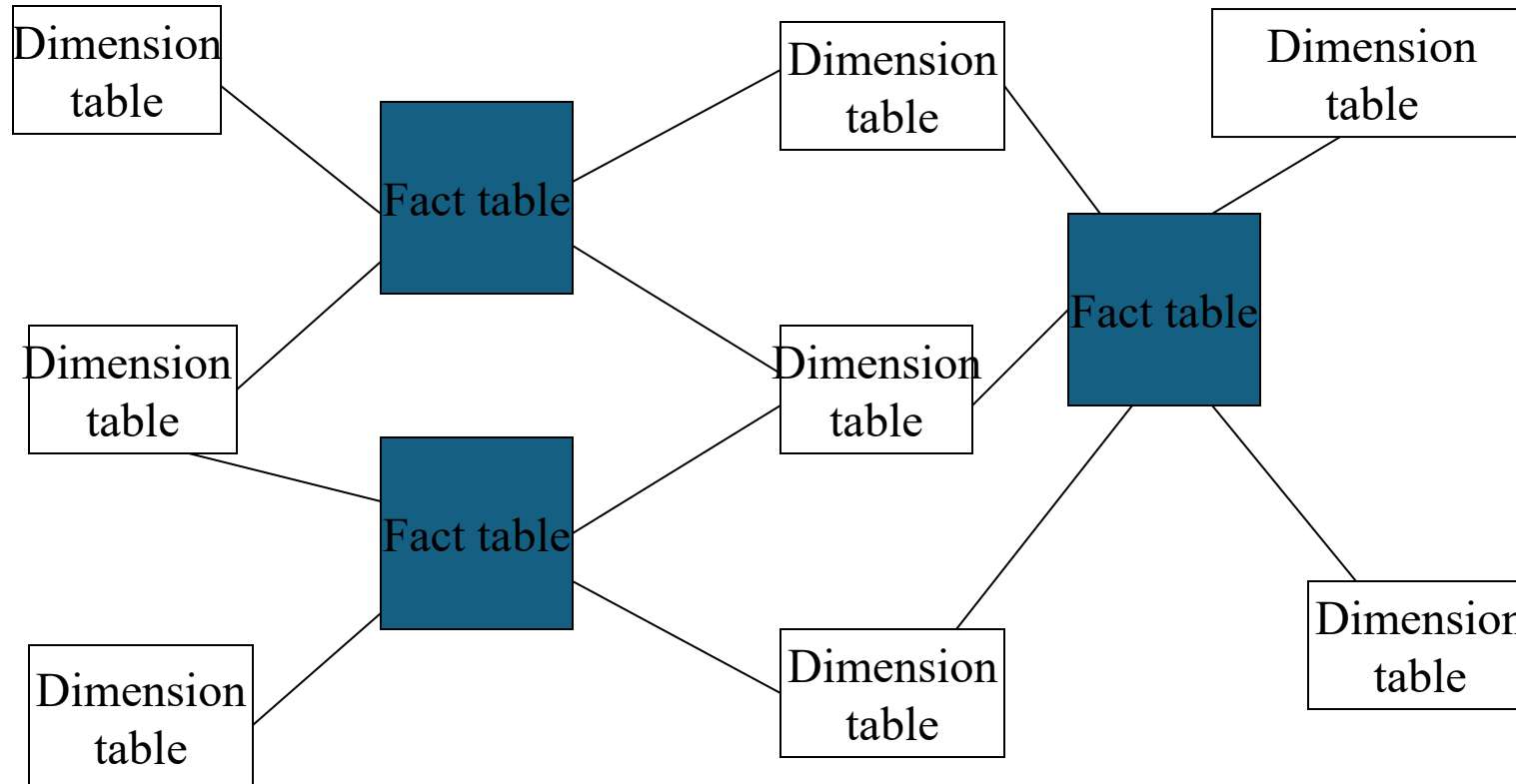
- Aggregate fact tables are summaries of the most granular data at higher levels along the dimension hierarchies.



The “Fact Constellation” Schema



Families of Stars



Data Warehousing

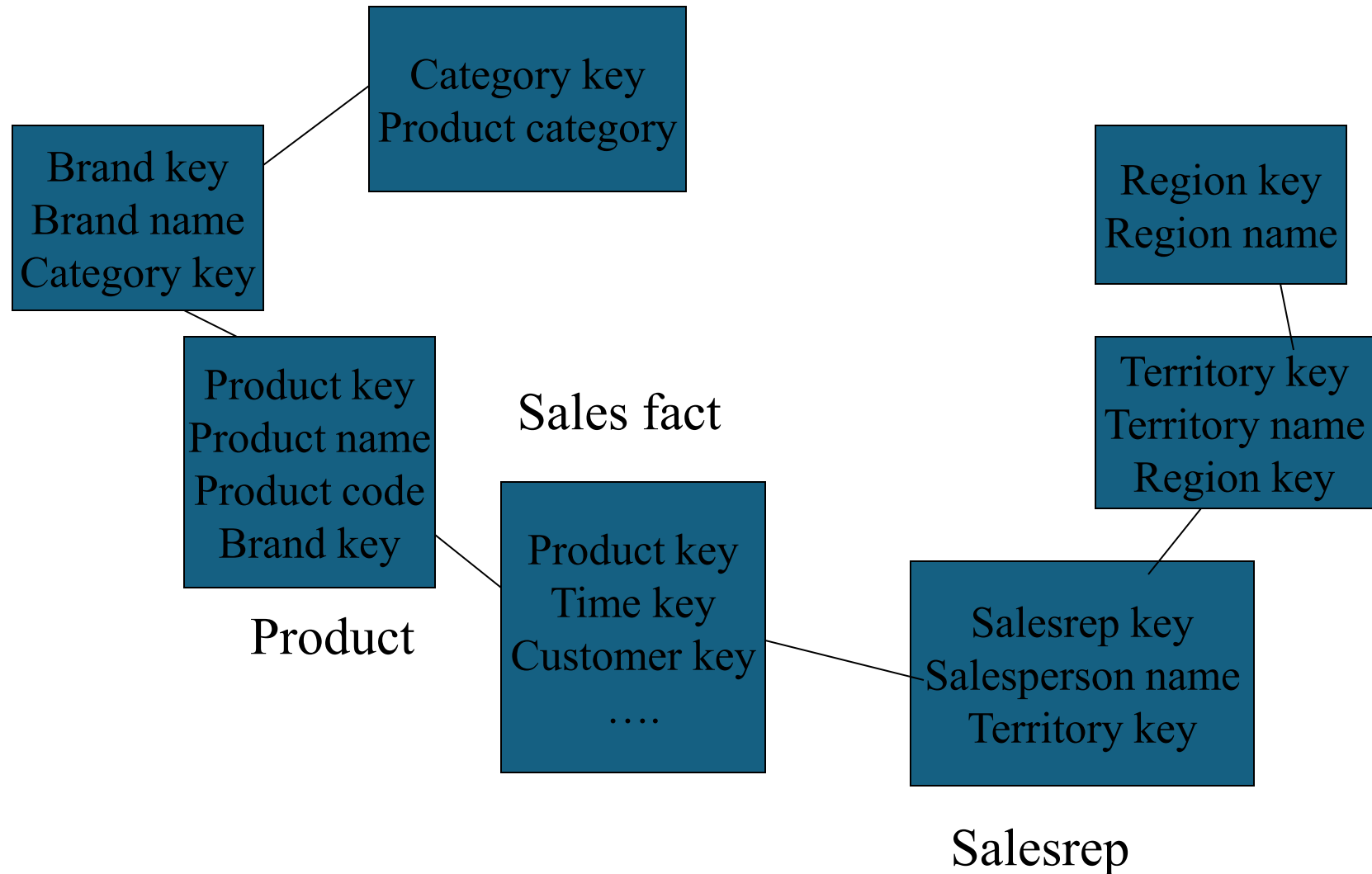
Lecture- 3: DW Modelling (Cont.)

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Snowflake Schema

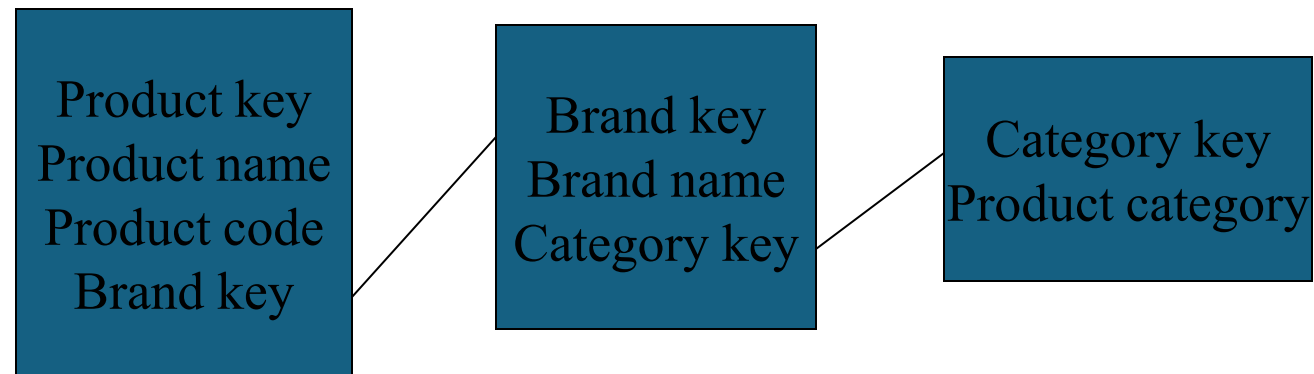
- **Snowflake schema** is a type of star schema but a more complex model.
- “**Snowflaking**” is a method of normalizing the dimension tables in a star schema.
- The normalization eliminates redundancy.
- The result is more complex queries and reduced query performance.

Sales: Snowflake Schema



Snowflaking

- The attributes with low cardinality in each original dimension table are removed to form separate tables (lookups)
- . These new tables are linked back to the original dimension table through artificial keys.



Snowflake Schema

- Advantages:
 - Small saving in storage space
 - Normalized structures are easier to update and maintain
- Disadvantages:
 - Schema less intuitive and end-users are put off by the complexity
 - Ability to browse through the contents difficult
 - Degrade query performance because of additional joins

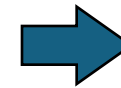
What is the Best Design?

- Performance benchmarking can be used to determine what is the best design.
- Snowflake schema: easier to maintain dimension tables when dimension tables are very large (reduce overall space). It is not generally recommended in a data warehouse environment.
- Star schema: more effective for data cube browsing (less joins): can affect performance.

Aggregates

- Add up amounts for day 1
- In SQL: `SELECT sum(amt) FROM SALE WHERE date = 1`

sale	prodId	storeId	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4

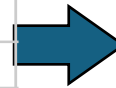


81

Aggregates

- Add up amounts by day
- In SQL: `SELECT date, sum(amt) FROM SALE GROUP BY date`

sale	prodId	storeId	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4

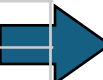


ans	date	sum
	1	81
	2	48

Another Example

- Add up amounts by day, product
- In SQL: `SELECT date, sum(amt) FROM SALE
GROUP BY date, prodId`

sale	prodId	storeId	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4



sale	prodId	date	amt
	p1	1	62
	p2	1	19
	p1	2	48

— rollup —→

← drill-down —

Aggregates

- Operators: sum, count, max, min, median, avg.
- “Having” clause
- Using dimension hierarchy
 - Average by region (within store)
 - Maximum by month (within date)

Data Cube

Fact table view:

sale	prodId	storeId	amt
	p1	s1	12
	p2	s1	11
	p1	s3	50
	p2	s2	8



Multi-dimensional cube:

	s1	s2	s3
p1	12		50
p2	11	8	

dimensions = 2

3-D Cube

Fact table view:

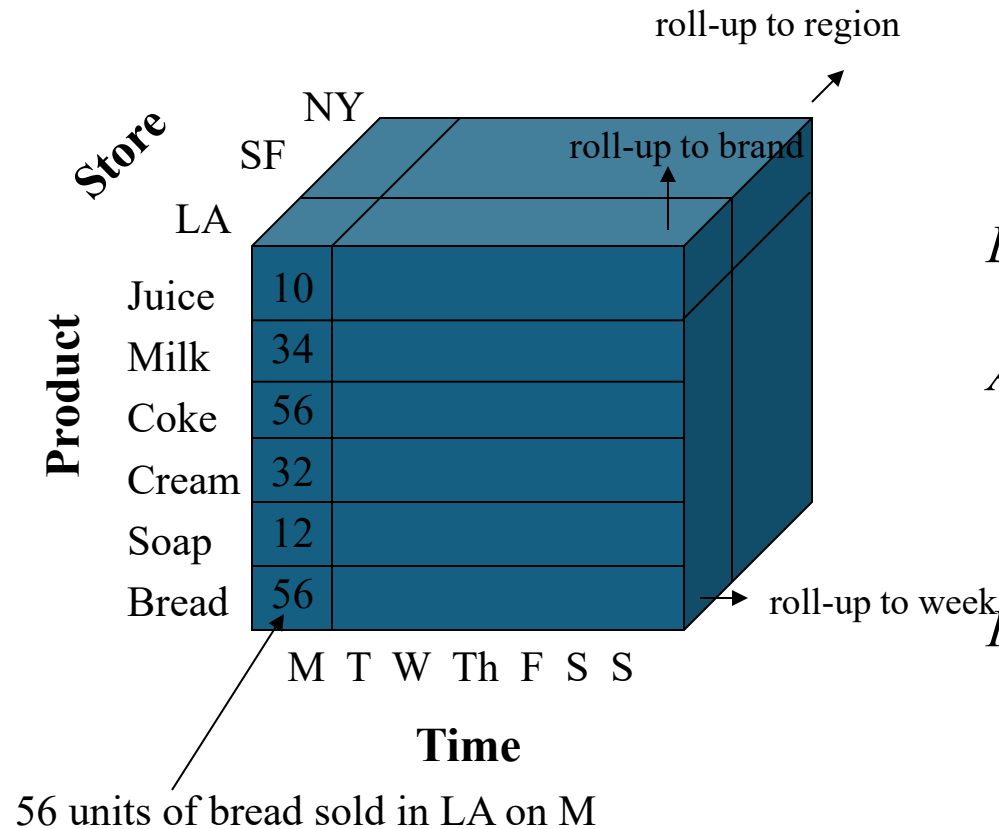
sale	prodId	storeId	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4

Multi-dimensional cube:

		s1	s2	s3
day 2	p1	44	4	
		s1	s2	s3
day 1	p1	12		50
	p2	11	8	

dimensions = 3

Example



Dimensions:

Time, Product, Store

Attributes:

Product (upc, price, ...)

Store ...

...

Hierarchies:

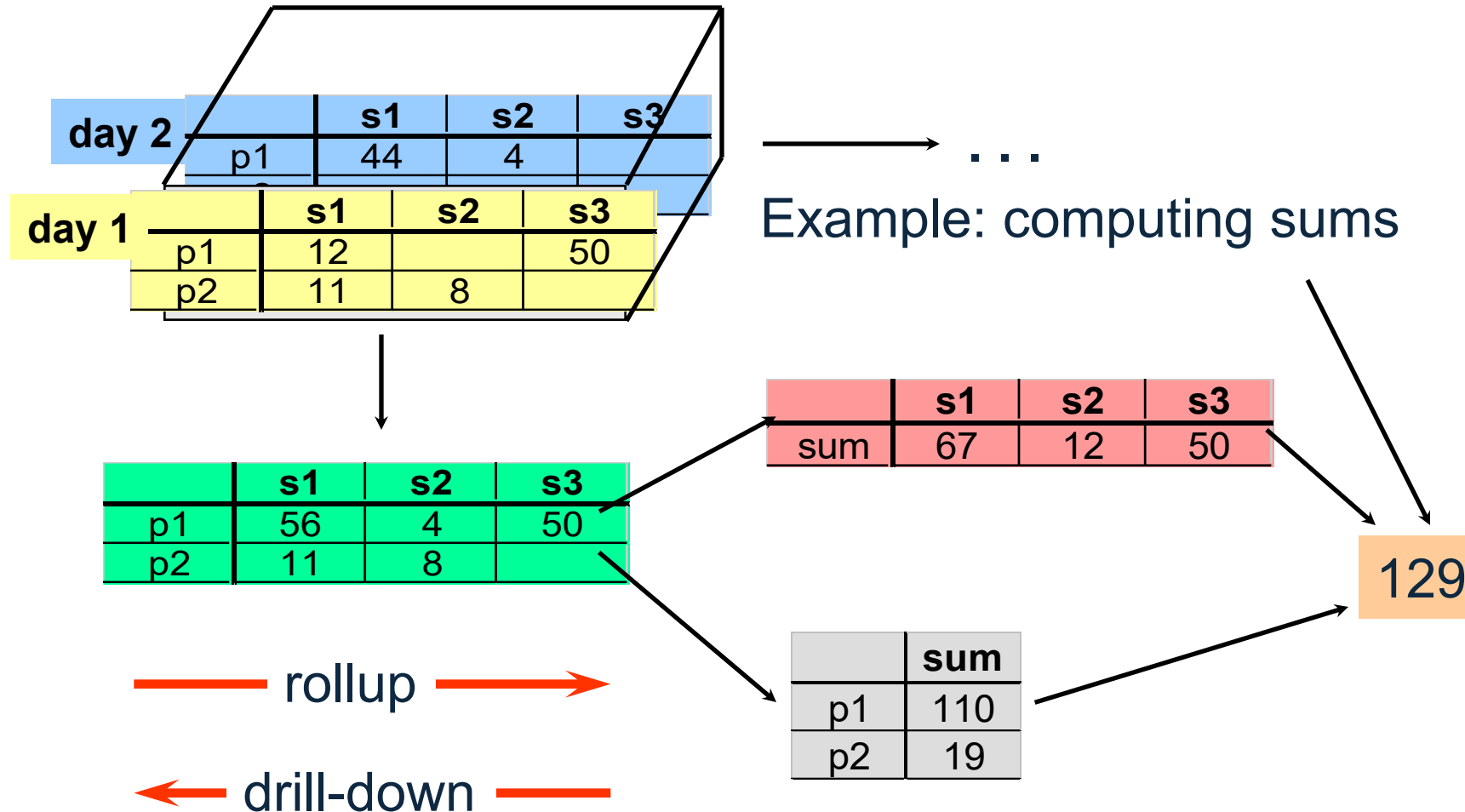
Product → Brand → ...

Day → Week → Quarter

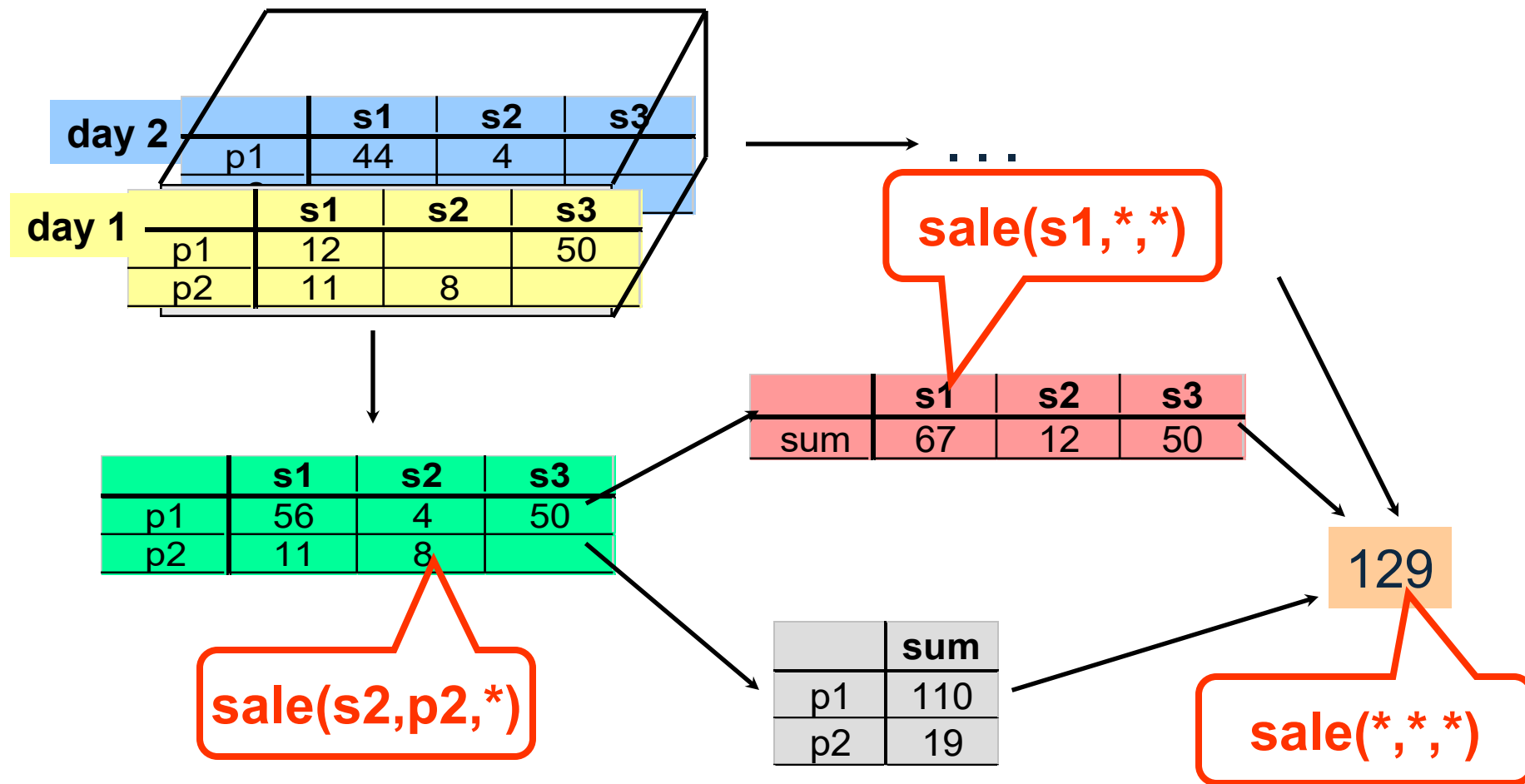
Store → Region →

Country

Cube Aggregation: Roll-up



Cube Operators for Roll-up



Extended Cube

The diagram illustrates an extended cube with two time periods, day 1 and day 2. Each period has a cube with dimensions p (product), s (supplier), and * (total). The data is as follows:

Day	p	s1	s2	s3	*
day 1	p1	12		50	62
	p2	11	8		19
	*	23	8	50	81
day 2	p1	44	4		48
	p2				
	*				

A red callout points to the p2 row in the day 2 cube, labeled **sale(*,p2,*)**.

Aggregation Using Hierarchies

day 2		s1	s2	s3
p1		44	4	
day 1		s1	s2	s3
p1		12		50
p2		11	8	

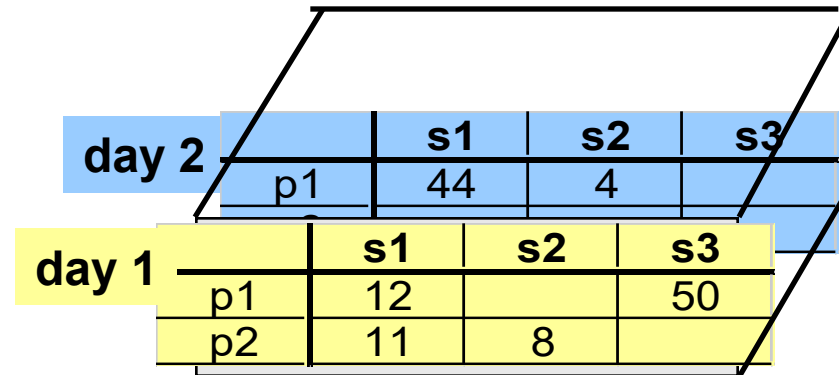


	region A	region B
p1	56	54
p2	11	8

store
|
region
|
country

(store s1 in Region A;
stores s2, s3 in Region B)

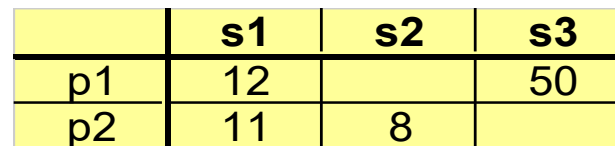
Slicing



day 2			
	s1	s2	s3
p1	44	4	

day 1			
	s1	s2	s3
p1	12		50
p2	11	8	

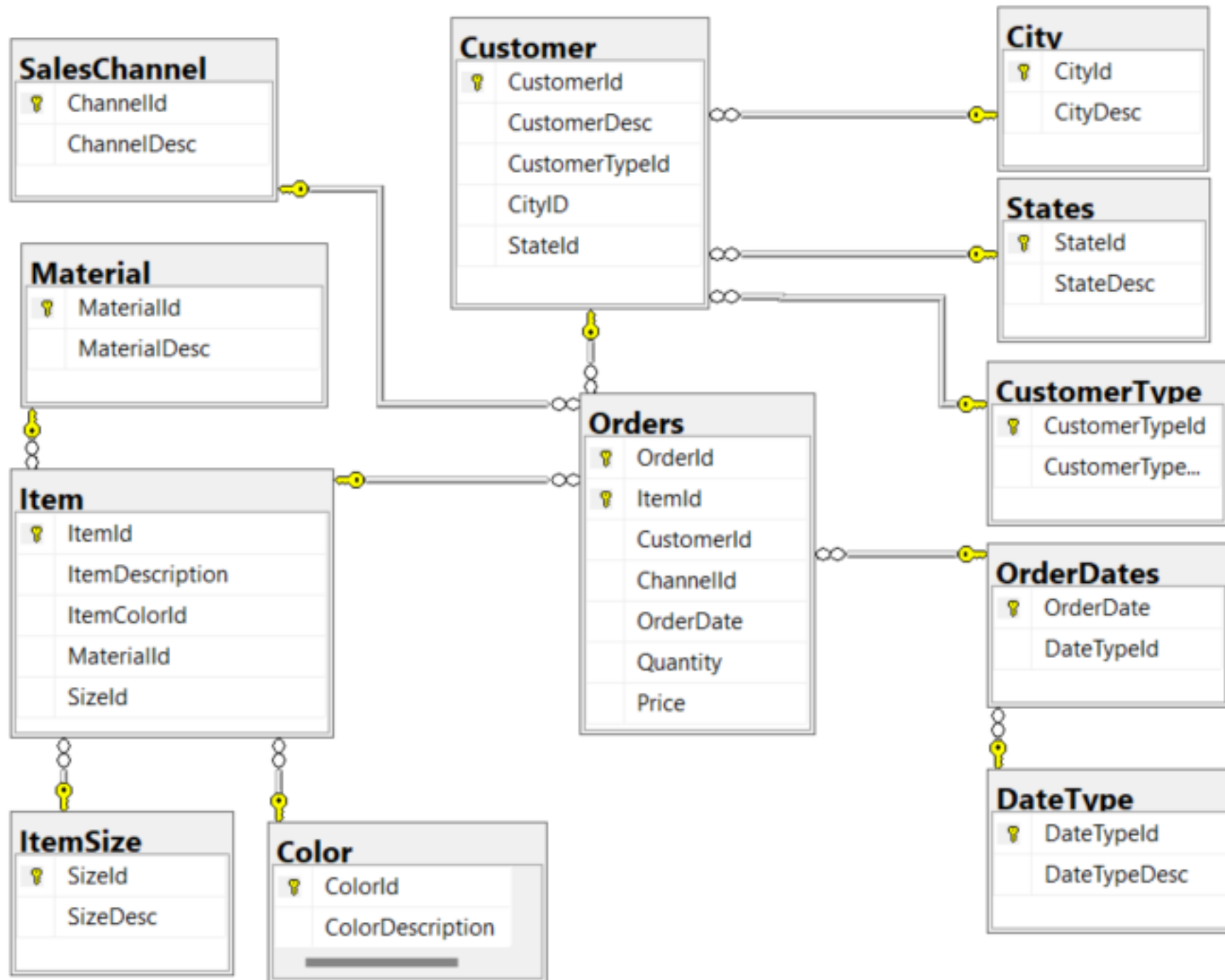
TIME = day 1



	s1	s2	s3
p1	12		50
p2	11	8	

Summary of Operations

- Aggregation (roll-up)
 - aggregate (summarize) data to the next higher dimension element
 - e.g., total sales by city, year → total sales by region, year
- Navigation to detailed data (drill-down)
- Selection (slice) defines a sub cube
 - e.g., sales where city = 'Gainesville' and date = '1/15/90'
- Calculation and ranking
 - e.g., top 3% of cities by average income
- Visualization operations (e.g., Pivot)
- Time functions
 - e.g., time average



Types of Facts/Measures

- The numeric measures in a fact table fall into three categories or types:
 - ❖ Fully Additive Measures/Facts.
 - ❖ Semi-Additive Measures/Facts.
 - ❖ Non-Additive Measures / Facts.
- The most flexible and useful facts are fully additive.
- Additive measures can be aggregated across any (or combination) of the dimensions associated with the fact table.
- An example of a fully additive measure is sales (purchases from a store).
- You can add hourly sales to get the sales for a day, week, month, quarter, or year. You can add sales across stores or regions.

Types of Facts/Measures

❖ Fully Additive Measures/Facts → Example

Numeric Features → {TransactionId, QuantitySold, UnitPrice, SalesAmount}

Text Features → DateTime, StoreName, ProductId

DateTime	StoreName	ProductId	TransactionId	QuantitySold	UnitPrice	SalesAmount
1/4/2022	Faisal	Prod-1	100	10	100	1000
1/4/2022	Faisal	Prod-1	200	20	100	2000
1/4/2022	Giza	Prod-2	300	30	200	6000
1/4/2022	Giza	Prod-3	400	20	200	4000
1/5/2022	Dokki	Prod-1	500	15	100	1500
1/5/2022	Haram	Prod-1	600	100	100	10000
1/5/2022	Haram	Prod-5	700	102	15	3030
1/6/2022	Asuit	Prod-4	800	12	100	1200
1/6/2022	Minya	Prod-2	1000	204	200	40800